

NAVIGATION SYSTEMS AND RADAR SERVICES

FULFILLS PA.VI.B, CA.VI.B, AI.II.H

Objective	
The student shall understand ground and satellite based navigation, radar services from ATC, and the functions and capabilities of transponders and ADSB. The student shall become familiar with utilizing these functions on local and cross country flights.	
Instructor Actions	Student Actions
- Explain ground-based navigation aids - Demonstrate ERAU <u>VOR simulator</u> - Explain satellite-based navigational systems - Demonstrate Garmin GPS 175 simulator - Discuss ATC radar services and flight following - Discuss functions and capabilities of ADSB and associated systems - Walk around aircraft and identify antennas and their function	- Take notes and participate in instructor's discussion - Practice self-quizzing and verifying expected depiction on VOR simulator - Read GPS <u>175 pilot's guide</u> and practice using iPad simulator - Understand limitations of navigation systems
Case Studies	Equipment
- Non-broadcasted traffic	- Additional resources - Computer/iPad - FAR/AIM - PHAK - White Board
Completion Standards	
The student shall explain identification and mitigation methods for aeromedical phenomena. The student shall demonstrate proficiency in incorporating navigation systems and radar services into local and cross country flights.	

ELEMENTS

1. Ground-Based Navigational Systems.....	1
1.1. VOR Theory and Regulations.....	1
1.2. The CDI and Use in Flight	1
1.3. Other Ground-Based Navigational Systems.....	1
2. Satellite-Based Navigational Systems	1
3. ATC Radar Services	1
3.1. Flight following.....	1
4. Transponders.....	1
4.1. Mode A.....	1
4.2. Mode C	1
4.3. Mode S / ADSB.....	1
5. Risks Involved.....	1
5.1. Ground-Based NAVAIDs.....	1
5.1. Satellite-Based NAVAIDs.....	1
6. Use of an EFB (ForeFlight/Garmin Pilot)	1

RESOURCES

FAA-S-ACS-6C Private Pilot ACS - Area VI Task B
FAA-S-ACS-7B Commercial Pilot ACS - Area VI Task B
FAA-S-ACS-25 CFI ACS - Area II Task H

FAA-H-8083-2 Risk Management Handbook
FAA-H-8083-3C Airplane Flying Handbook
FAA-H-8083-9 Aviation Instructors Handbook
FAA-H-8083-25C PHAK Chapter 16: Navigation

AC 91-78A – Use of Electronic Flight Bags

1. GROUND-BASED NAVIGATIONAL SYSTEMS

1.1. VOR Theory and Regulations

VOR lighthouse example

VOR inspection regulations (91.171)

1.2. The CDI and Use in Flight

1.3. Other Ground-Based Navigational Systems

DME (specify slant range)

NDB (outline bearing pointer)

2. SATELLITE-BASED NAVIGATIONAL SYSTEMS

GPS and RAIM, WAAS (SBAS), Databases

Identification of Antennas

3. ATC RADAR SERVICES

3.1. Flight following

4. TRANSPONDERS

4.1. Mode A

Most basic, broadcasts squawk code

4.2. Mode C

Also broadcasts pressure altitude

4.3. Mode S / ADSB

4.4.

5. RISKS INVOLVED

5.1. Ground-Based NAVAIDs

Reverse sensing

5.1. Satellite-Based NAVAIDs

Loss of satellite signal

6. USE OF AN EFB (FOREFLIGHT/GARMIN PILOT)

Per 91-78A, the FAA has legalized the use of a current chart on an EFB as a suitable replacement for paper charts.